Arshdeep Singh

+91 77386 69323
work@arshwaraich.com
arshw.com

Summary

Software engineer at Stripe, 5+ years in authentication, risk, and investments. Impact across Stripe auth and API keys (100M+ developers), the HSBC–RBC acquisition ($1B+), and Canada's first-to-market FHSA.

Work Experience

May 2025 – Present **Software Engineer**, *Stripe*.

Marketing Technology, Bengaluru, KA, Aug 2026 – Present

- Rebuilt Marketing's LLM help bot around skills with confidence-based human handoff; a Claude Code eval loop lifted solve rate from 12% to 88%; it now answers 44% of requests.
- Automated Marketo lead backfills with a crash-resumable Airflow + S3 queue that stays within API quota, ending a manual process that had caused 3 incidents.

User Auth Experience (auth & identity security), Toronto, ON, May 2025 – Aug 2026

- DRI for Email OTP as a new Stripe Connect auth factor with a per-user/country/domain SMS kill switch; launched email-only login in 8 high-fraud countries (~$96K/month in SMS savings).
- DRI for risk scoring on Connect (26M+ users) and API keys (100M+ developers); cut consumer runtime from ~1 day to seconds by fixing concurrency and MongoDB lock contention. Flagged 250K+ anomalies at <10% false positives; 754 merchants rotated keys.
- Sole DRI for a bank-account update-code flow, shipped ahead of schedule to 7,672 platforms.
- Sole frontend owner for restricted-key onboarding; new-merchant adoption rose 12% to ~60%.
- Built LLM evals and the approval page for MCP Human-in-the-Loop, shipped for refunds.

Aug 2023 – May 2025 **Senior Software Developer**, *Royal Bank of Canada*, Toronto, ON.

- Integrated HSBC clients into RBC Direct Investing for the largest acquisition in Canadian banking history, adding $1B+ to AUM, delivered on time.
- Redesigned the Direct Investing homepage and helped enable foreign-market trading.

Sep 2021 – Aug 2023 **Software Developer**, *Royal Bank of Canada*, Toronto, ON.

- Led front-end changes for Canada's first-to-market First Home Savings Account (FHSA) on RBC InvestEase, delivering within 4 months and enabling $100M in Q1 investments.
- Developed bulk account approvals, reducing manual processing by 40%.

Jan 2021 – Apr 2021 **Software Developer Co-Op**, *Royal Bank of Canada*, Toronto, ON.

- Built Java Spring batch migration tools for legacy system decommissioning; recognised by RBC's Group Head of Technology & Operations for exceptional performance.

May 2020 – Aug 2020 **Software Developer Co-Op**, *Portable Intelligence Inc.*, Markham, ON.

- Built a UDP integration layer for the Artemis computer vision system, opening a new revenue stream; automated updates ended on-site client visits.

Education

Jan 2018 – Aug 2021 **Honours Bachelor of Technology – Software Development (BSD)**, *Seneca College*, Toronto, ON.

- GPA: 4.0
- President's Honour List: Fall 2018 - Summer 2021

Skills

Languages	Ruby · TypeScript · C# · HTML/CSS · Java · Python · Swift
Frameworks	.NET · ReactJS · NextJS · React Native · SwiftUI · NodeJS
Data	MongoDB · PostgreSQL · DynamoDB · Redis · SQLite · S3 · Airflow · Parquet
AI	Claude Code agents & skills · LLM evals (Braintrust) · MCP · Prompt optimization · RAG · LangChain
Other	Authentication (OTP, WebAuthn, passkeys) · Risk scoring · RESTful APIs · CI/CD · Grafana · Figma · AWS Serverless

Projects

wallet-mcp, *Python, MCP, Starlette, Pillow, SQLite, x402.*
An MCP server and REST API that turns tickets, boarding passes, and loyalty cards into signed Apple Wallet passes from Claude, ChatGPT, or Codex, with live pass updates over APNs and x402 USDC micropayments past the free tier. ~2,700 passes signed for ~800 distinct callers in its first 8 weeks; listed on the official MCP Registry. · walletmcppass.com

Clarity Softphone, *Swift, SwiftUI, Linphone SDK, CallKit.*
A native iOS SIP softphone supporting UDP/TCP/TLS transports, CallKit-integrated call UX, and DTMF. Shipped to the App Store. · github.com/arshwaraich/softphone

Sun Timezone, *Swift, SwiftUI, Figma, NextJS.*
A web, iOS, and macOS app visualising multiple timezones along a sun-arc chart with a scrubbing time dial, plus home-screen widgets. Ported from an earlier NextJS web app (arshwaraich.github.io/time-zone) to native iOS using LLM coding agents; shipped to the App Store. · github.com/arshwaraich/Timezone

Awards

One RBC Team Award, *2023 Q3, 2024 Q1, 2024 Q3.*
Recognized for work in the HSBC acquisition, FHSA delivery and the Direct Investing Home redesign project, improving cross-functional collaboration and achieving key milestones on time.
Royal Bank of Canada

Certificate of Recognition, *April 2020.*
In recognition of contributions to innovative research and entrepreneurial activities at Seneca. Presented at the Annual Applied Research, Innovation & Entrepreneurship Showcase.
Seneca Applied Research

RBC AmpHacks Top 5, *October 2020.*
For creating a video-sharing app for seniors called SocialDisco.
RBC AmpHacks